Musterlösung Serie 19

Eigenvectors, Eigenvalues

1. In each of the following cases, let T_i be the endomorphism of $\mathbb{R}^2$ which is represented by the matrix A_i in the standard ordered basis for $\mathbb{R}^2$, and let U_i be the endomorphism of $\mathbb{C}^2$ represented by A_i in the standard ordered basis. Find the characteristic polynomial for T_i and that for U_i, find the eigenvalues of each endomorphism, and for each such eigenvalue find a basis for the corresponding space of eigenvectors.

$$A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 2 & 3 \\ -1 & 1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Solution: We have

$$\mathrm{char}_{T_1}(X) = \mathrm{char}_{U_1}(X) = X(X-1).$$

Hence both T_1 and U_1 have real eigenvalues 0 and 1. We compute that

$$\mathrm{Eig}_{T_1}(0) = \left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle_{\mathbb{R}}, \quad \mathrm{Eig}_{T_1}(1) = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle_{\mathbb{R}}$$

and similarly for U_1 but taking the span over $\mathbb{C}$ instead of $\mathbb{R}$.

In the second case we have

$$\mathrm{char}_{T_2}(X) = X^2 - 3X + 5 \in \mathbb{R}[X].$$

This polynomial doesn't split into linear factors in $\mathbb{R}[X]$ hence the endomorphism T_2 does not have any real eigenvalues. However, if we now consider $U_2 \in \mathrm{End}(\mathbb{C}^2)$, $\mathrm{char}_{U_2}(X) \in \mathbb{C}[X]$ splits into linear factors and takes $\lambda_1 = \frac{1}{2}(3 + i\sqrt{11})$ and $\lambda_2 = \frac{1}{2}(3 - i\sqrt{11})$ for roots.

We use a handy trick to easily compute eigenvectors:

Eigenvector trick for 2×2 matrices. *Let A be a 2×2 matrix, and let λ be a (real or complex) eigenvalue of A. Then*

$$A - \lambda I_2 = \begin{pmatrix} \alpha & \beta \\ * & * \end{pmatrix} \implies \begin{pmatrix} -\beta \\ \alpha \end{pmatrix} \text{ is an eigenvector with eigenvalue } \lambda,$$

assuming the first row of $A - \lambda I_2$ is non-zero.

Explanation. Indeed since λ is an eigenvalue, $A - \lambda I_2$ has nontrivial kernel. It follows that its rows are collinear, i.e. that the second row is a complex multiple of the first one:

$$\begin{pmatrix} \alpha & \beta \\ * & * \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \gamma\alpha & \gamma\beta \end{pmatrix}, \qquad \text{for some } \gamma \in \mathbb{C}.$$

So $\left(\begin{smallmatrix} -\beta \\ \alpha \end{smallmatrix} \right)$ is an obvious element of that kernel.

Applying this to A_2 and its eigenvalue λ_1, we get

$$A_2 - \lambda_1 I_2 = \begin{pmatrix} 2 - \lambda_1 & 3 \\ * & * \end{pmatrix}$$

which implies that $\mathrm{Eig}_{U_2}(\lambda_1)$ is generated by

$$\begin{pmatrix} -3 \\ 2 - \lambda_1 \end{pmatrix} = \begin{pmatrix} -3 \\ \frac{1}{2}(1 - i\sqrt{11}) \end{pmatrix}.$$

Similarly, we recover that $\mathrm{Eig}_{U_2}(\lambda_2)$ is generated by

$$\begin{pmatrix} -3 \\ \frac{1}{2}(1 + i\sqrt{11}). \end{pmatrix}$$

We treat T_3 and U_3 similarly as T_1 and U_1. Their characteristic polynomial splits into linear factors in $\mathbb{R}[X]$:

$$\mathrm{char}_{T_3}(X) = \mathrm{char}_{U_3}(X) = X(X-2).$$

The eigenvalues are therefore 0 and 2. We find that

$$\mathrm{Eig}_{T_3}(0) = \left\langle \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\rangle_{\mathbb{R}}, \quad \mathrm{Eig}_{T_3}(2) = \left\langle \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\rangle_{\mathbb{R}}.$$

Similarly for U_3 over $\mathbb{C}$.

2. Let K be a field and let V be a finite-dimensional vector space over K. Suppose that $T \in \mathrm{End}(V)$ is invertible. Prove that $\mathrm{Eig}_T(\lambda) = \mathrm{Eig}_{T^{-1}}(1/\lambda)$ for every $\lambda \in K^*$.

   Lösung: This easily follows from the definition:

$$\begin{aligned} \mathrm{Eig}_T(\lambda) &= \{v \in V \mid Tv = \lambda v\} \\ &= \{w \in V \mid w = \lambda T^{-1} w\} \\ &= \left\{ w \in V \mid \frac{1}{\lambda} w = T^{-1} w \right\} \\ &= \mathrm{Eig}_{T^{-1}}\left(\frac{1}{\lambda}\right). \end{aligned}$$

Remark. We didn't use the assumption that V is finite-dimensional. In fact, this statement also holds in infinite-dimensional vector spaces by the same proof.

3. Consider the space $C^\infty(\mathbb{R})$ of smooth functions over $\mathbb{R}$ and the map

$$\begin{array}{rccc} T: & C^\infty(\mathbb{R}) & \to & C^\infty(\mathbb{R}) \\ & f & \mapsto & f' \end{array}$$

Find the eigenvalues and the corresponding eigenfunctions (this is a synonym for eigenvectors when working on a space whose elements are functions) of T.

Solution: For $\lambda \in K$, we can get an idea of the solution by solving the linear ordinary differential equation

$$\frac{\mathrm{d}\, f(x)}{\mathrm{d}\, x} = \lambda f(x).$$

We have

$$\begin{aligned} & \frac{1}{f(x)}\frac{\mathrm{d}\, f(x)}{\mathrm{d}\, x} = \lambda \\ \implies \quad & \int \frac{1}{f(x)}\frac{\mathrm{d}\, f(x)}{\mathrm{d}\, x}\,\mathrm{d}\, x = \int \lambda\,\mathrm{d}\, x \end{aligned}$$

Substituting u for $f(x)$ on the left-hand side, we obtain $\mathrm{d}\, u = \frac{\mathrm{d}\, f(x)}{\mathrm{d}\, x}\,\mathrm{d}\, x$

$$\begin{aligned} & \int \frac{1}{u}\,\mathrm{d}\, u = \lambda x + C, \quad C \in K \\ \implies \quad & \log(u) = \lambda x + C \\ \implies \quad & \log(f(x)) = \lambda x + C \\ \implies \quad & f(x) = e^{\lambda x + C}. \end{aligned}$$

Hence, the family $\{f(x) = f(0)e^{\lambda x} \mid \lambda \in K\}$ is a set of eigenfunctions of T and, at least formally, eigenfunctions should have the form $f(x) = f(0)e^{\lambda x}$ for $\lambda \in K$.

To check that these are the only possible solutions, we use the following trick: consider a solution f_0 of the differential equation $\frac{\mathrm{d}\, f(x)}{\mathrm{d}\, x} = \lambda f(x)$ and define the modified function

$$g_0(x) = e^{-\lambda x} f_0(x).$$

Now,

$$\begin{aligned} \frac{\mathrm{d}\, g_0}{\mathrm{d}\, x}(x) &= -\lambda e^{-\lambda x} f_0(x) + \lambda e^{-\lambda x} f_0(x) \\ &= 0. \end{aligned}$$

We deduce that $g_0(x)$ is constant, hence for all $x \in \mathbb{R}$,

$$g_0(x) = g_0(0) = f_0(0) \Leftrightarrow f_0(x) = f_0(0)e^{\lambda x}.$$

4. Let $K = \mathbb{R}$, show that K^∞ does not admit any countable basis.

   Hint: Use the fact that pairwise distinct eigenvalues correspond to a set of linearly independent eigenvectors.

   Solution: Let $\lambda \in K$ and consider the sequence $L_\lambda := (1, \lambda, \lambda^2, \lambda^3, \dots)$. Apply the shift operator

   $$\begin{array}{rccc} S : & K^\infty & \to & K^\infty \\ & (a_0, a_1, a_2, a_3, \dots) & \mapsto & (a_1, a_2, a_3, \dots) \end{array}$$

   to L_λ and observe that (λ, L_λ) is an eigenvalue-eigenvector pair for S. Hence S admits an uncountable number of eigenvalues since each $\lambda \in K$ is one. Moreover, as seen in the lectures, since these eigenvalues are distinct, the set $\{L_\lambda \mid \lambda \in K\}$ is linearly independent. Using Serie 7 exercise 5, we conclude that K^∞ does not admit any countable basis.

5. Let K be a field and let V be an n-dimensional vector space over K ($n > 0$).

   (a) Let T be a diagonalizable endomorphism of V with (not necessarily distinct) eigenvalues λ_i for $1 \leqslant i \leqslant n$. Show that

   $$\mathrm{Tr}(T) = \sum_{i=1}^n \lambda_i \quad \text{and that} \quad \det(T) = \prod_{i=1}^n \lambda_i.$$

   For $0 \leqslant k \leqslant n$, let c_k be the coefficient of x^k in the characteristic polynomial of T. Give a formula for c_k in terms of the eigenvalues of T.

   (b) Let $B \in M_{2\times 2}(\mathbb{R})$ be diagonalizable with $\mathrm{Tr}(B) = 0$. Show that $\det(B) \leqslant 0$.

   Solution:

   (a) We know that we can factor the characteristic polynomial of T as

   $$\mathrm{char}_T(X) = \prod_{i=1}^n (\lambda_i - X).$$

   It follows from this formula that

   $$c_k = (-1)^k \sum_{\substack{\{i_1, i_2, \dots, i_{n-k}\} \in \{1,\dots,n\}^{n-k} \\ i_r \neq i_s \text{ for } r \neq s}} \lambda_{i_1} \lambda_{i_2} \cdots \lambda_{i_{n-k}}$$

   In particular,

   $$c_0 = \prod_{i=1}^n \lambda_i$$

   $$c_{n-1} = (-1)^{n-1} \sum_{i=1}^n \lambda_i.$$

You have seen in the lectures that $c_0 = \det(T)$ and $c_{n-1} = (-1)^{n-1}\operatorname{Tr}(T)$. This shows what we wanted.

Aliter: Since T is diagonalizable, by definition there exists a basis $\mathcal{B}$ of V such that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}.$$

You have seen in the lectures that the trace and the determinant do not depend of the choice of basis, hence by a direct computation we have

$$\operatorname{Tr}(T) = \operatorname{Tr}([T]_{\mathcal{B}}^{\mathcal{B}}) = \sum_{i=1}^{n} \lambda_i \quad \text{and} \quad \det(T) = \det([T]_{\mathcal{B}}^{\mathcal{B}}) = \prod_{i=1}^{n} \lambda_i.$$

(b) Since B is diagonalizable, we can use (a) and observe that

$$\operatorname{Tr}(B) = 0 \Longleftrightarrow \lambda_1 + \lambda_2 = 0 \Longleftrightarrow \lambda_1 = -\lambda_2,$$

where λ_i denote the eigenvalues of B. Hence, using (a) again,

$$\det(B) = -\lambda_1^2 \leqslant 0$$

since $\lambda_1^2 \geqslant 0$.